

Mohamed Elkholy

Full-Stack Engineer · Independent Contractor

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SUMMARY

Full-stack engineer with **8+ years** building **TypeScript**, **Node.js**, and **React** applications — **100%** Job Success Score and **7,600+** hours delivered on Upwork, plus full-time roles. Developing scalable, cloud-native applications on **GCP** and **AWS**.

EXPERIENCE

Full-Stack Engineer, Freelance · Remote

2014 – Present

- Delivered end-to-end web apps for startups and SMBs—from microsites/MVPs to large-scale applications.
- Designed serverless architectures on AWS, GCP, and Firebase for scalable, low-ops, cost-efficient delivery.
- Built React/Angular UIs and Node.js services; integrated third-party APIs, auth, payments, AI, and webhook flows.
- Provided technical consulting: architecture/code reviews, tool selection, performance audits, and workflow improvements.
- **2023** — Checkout workflow revamp on React/Node.js (GCP) with Stripe integration, increasing conversion by 12%.
- **2021** — Firebase/Cloud Functions; optimized indexes, caching, and queries, reducing cloud costs by 28%.
- **2020** — React/Next.js + Shopify (GraphQL API); delivered a storefront MVP in 6 weeks.

Full-Stack Engineer (Node.js/TypeScript), Mishe · Contract · United States (Remote)

2024 – 2025

- Modernized a large-scale Firebase codebase (web/Cloud Functions) to LTS, improving performance and release stability.
- Migrated the internal dashboard to MUI; built a reusable component library, improving UX and speeding feature delivery.
- Structured Firestore data with a schema analyzer and dashboard-driven migration runner, improving data integrity.
- Built Stripe operations dashboards with reconciliation and audit trails to reduce discrepancies and support compliance.
- Set up CI/CD with GitHub Actions + Cloud Pub/Sub: PR checks, preview deploys, and in-dashboard deployment logs.
- Improved quality and reliability with automated unit and E2E tests on PRs, structured logging, observability, caching and load testing, catching regressions earlier and keeping latency low.

Full-Stack Engineer (Node.js/TypeScript), JOKR · Full-time · United States (Remote)

2021 – 2024

- Owned NestJS GraphQL microservices end-to-end, standardized schemas and versioned releases, reducing regressions.
- Built internal React/MUI dashboards and reusable component libraries, streamlining operations and speeding delivery.
- Integrated an ML-driven recommendations service using user signals, boosting product CTR/AOV, driving revenue growth.
- Improved performance with DataLoader caching and batching, efficient pagination, and media delivery optimizations; validated via targeted load testing to sustain thousands of requests/min at low p95 latency.
- Achieved 90%+ test coverage (Jest, Cypress); implemented Datadog observability and runbooks to reduce MTTR.
- Set up CI/CD with the DevOps team using GitHub Actions and Terraform: Docker/K8s/Helm deploys to staging/production.
- Adopted AI-assisted dev workflows for refactors, tests, and docs, speeding PRs and improving documentation coverage.

"Although he started as a freelancer, Mohamed integrated seamlessly with our team, delivered **full-stack features** end-to-end, worked proactively **across time zones**, and shipped **well-tested, reliable code** while supporting other engineers."

— Ben Chen, VP of Engineering, JOKR

SKILLS

Soft Skills & Practices: Continuous learning, Problem-solving, Decision-making, Systems thinking, Communication, Organization, Code reviews, Test-driven development (TDD), Clean architecture, Documentation

Tech stack: Node.js, TypeScript, NestJS, Express.js, Better Auth, MongoDB, Redis, PostgreSQL, Apollo GraphQL, Google Cloud, Amazon AWS, Firebase, React, Angular, Vite, Material UI, RxJS, Stripe.js, Auth0, Vitest, Playwright, Docker, GitHub Actions

Also used: PHP, Python, Shopify, MySQL, Elasticsearch, Vercel Platform, JavaScript (ES6+), Gatsby, Next.js, Tailwind CSS, Redux, Webpack, Jest, K6, Cypress, Kubernetes, Terraform

EDUCATION

Formal degree with self-directed CS foundation and ongoing coursework

Professional Development · Selected Online Coursework 2015 – Present

AI Agents with MongoDB (2024) · Eloquent JavaScript (2024) · GraphQL Essentials (2020) · Advanced React (2018)

HarvardX (edX) – Online program · CS50x: Introduction to Computer Science 2016 – 2017

C · Python · SQL · Algorithms · Cryptography · Data Structures · Security

Misr University for Science and Technology · B.Sc. in Physical Therapy 2010 – 2016

FEATURED PROJECTS

JOKR — AI-powered grocery shopping app

- **Stack:** TypeScript/Node.js (NestJS), GraphQL (Apollo Federation), React/MUI, GCP, Docker/Kubernetes, Jest/Cypress.
- Delivered federated GraphQL microservices powering the iOS/Android app.
- Built internal operations dashboards and CI/CD pipelines.
- Integrated AI-driven product recommendations with guardrails.
- **Case study:** <https://mohatt.dev/projects/jokr>

Chatbot — Open source AI chatbot

- **Stack:** Next.js, Vercel AI SDK, PostgreSQL, Upstash Vector, Vercel AI Gateway.
- Streaming chat UI with Markdown rendering and rich AI elements.
- File uploads + RAG with vector indexing and file search/read tools.
- Multi-model support via Vercel AI Gateway.
- **Live demo:** <https://chat.mohatt.dev>